4.2.2 E.m.f. and p.d

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	potential difference (p.d.); the unit volt	
(b)	electromotive force (e.m.f.) of a source such as a cell or a power supply	The ASE guide 'Signs symbols and systematics' details E as the symbol for e.m.f.
(c)	distinction between e.m.f. and p.d. in terms of energy transfer	This should not be confused with the symbols for energy, electric field strength and for Young modulus which are also denoted as E.
(d)	energy transfer; $W = VQ$; $W = EQ$.	

(e) energy transfer $eV = \frac{1}{2}mv^2$ for electrons and other charged particles.